

Electronics & Communication Systems

Assignment Practical Tips



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Template for Process

Combinational

```
-- Comb Process
comb_process_label: process(a, b, [...])
-- Sensitivity list shall include all the signals
-- that are read by the process (the inputs of the
-- comb network).
begin
  c <= '0';
  d <= (others => '0');
  -- It is considered good practice to assign a
  -- default value to comb outputs. This prevents
  -- the inferring of latches!

  -- Comb function
  if ([...]) then
    c <= [...];
    d <= [...];
  end if;
end process;
```

Template for Process

Sequential

```
-- Seq Process, async reset, active low
seq_process_label: process(clk, a_rst_n)
begin
    if (a_rst_n = '0') then
        -- reset values for all signals
        -- assigned within the process
        a <= '0';
        b <= (others => '0');
    elsif (rising_edge(clk)) then
        -- Process Body
        a <= [...];
        b <= [...];
    end if;
end process;
```

```
-- Seq Process, sync reset, active high
seq_process_label: process(clk)
begin
    if (rising_edge(clk)) then
        if (rst = '1') then
            -- reset values for all signals
            -- assigned within the process
            a <= '0';
            b <= (others => '0');
        else
            -- Process Body
            a <= [...];
            b <= [...];
        end if;
    end if;
end process;
```

Instance of components

- Component declaration:
 - Component declaration is basically the copy of the original entity (both generics and ports)
- Component instance:
 - When instancing a component, remember to specify:
 - «generic map» (if generics are used)
 - «port map»
 - Generics and ports names on the **left** side of assignments must be the same of the original entity
 - Generics, ports and signals on the **right** side of assignment can have the same name of the ones on the left (e.g: clk)
 - Multiple instances of the same component are allowed, provided that they have different labels

Use of Variables

- Know the difference between variables and signals (See 1st set of slides).
- Variables may be necessary when working with loops and indexing.
- Variables can be used to store temporary values and improve reading of process VHDL code.
- If you use them, you will be asked to motivate why you used a variable instead of a signal.

Use of Constants

- Beware, do not use initialised signals as a constant. Vivado may prune them if they only keep their default value.
- Tricky, it will work without any problem or warning in simulation, but may give issues during Synthesis
- Use Constants!

```
-- Wrong use
signal CONSTANT_VALUE : integer := 10;
signal CONSTANT_VEC   : std_logic_vector(N-1 downto 0)
                        := (N-3 => '1', others => '0');

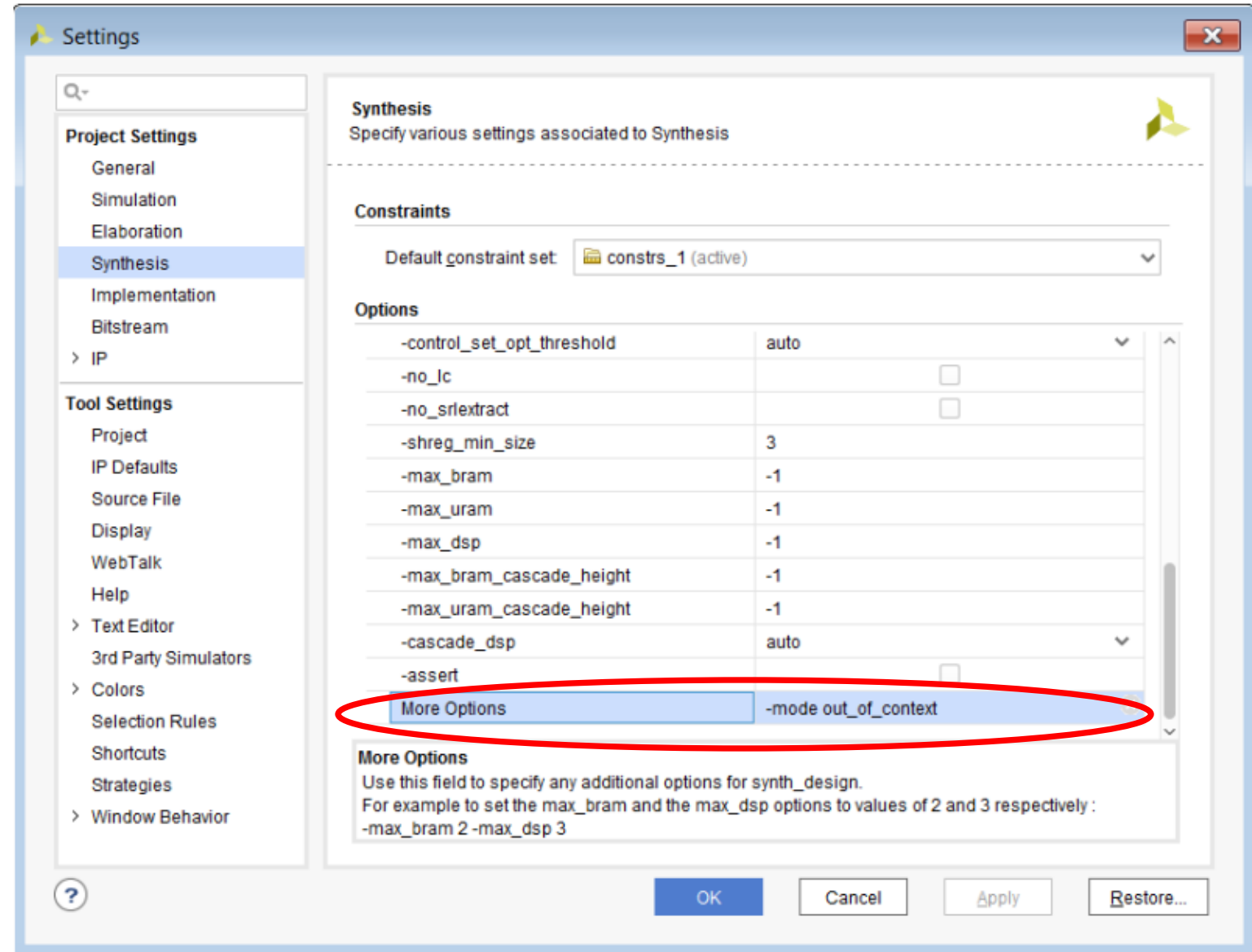
-- Correct use
constant CONSTANT_VALUE : integer := 10;
constant CONSTANT_VEC   : std_logic_vector(N-1 downto 0)
                        := (N-3 => '1', others => '0');
```

Signal Initialisation

- You MUST initialise all the signals used as input in Testbenches.
- You MUST NOT initialise any other signals.
- Tricky! You may apparently fix bugs by initialising signals in your HDL sources, but you are just hiding the problem, which will be back in FPGA operation.

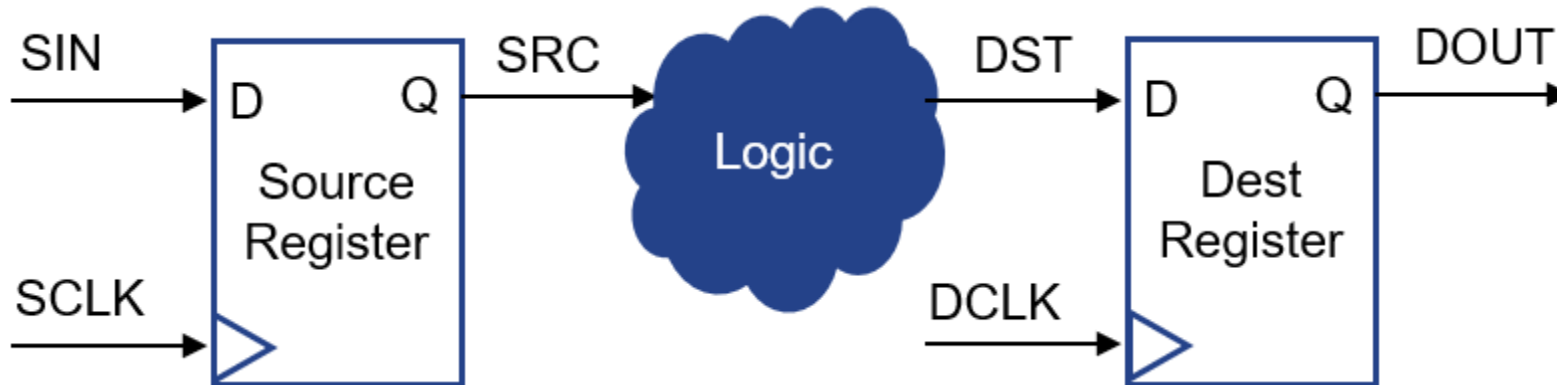
Out-of-context Synthesis mode

- To perform also the Implementation of your circuit, you do not want to map the I/O of your circuit to the FPGA physical pins.
- You can set Vivado to skip that part and perform the circuit implementation within the inner part of the FPGA.
- Add **-mode out_of_context** within the synthesis options.



I/O Registers

- Vivado evaluates timing only for “Register - Logic - Register” paths.
- In order for Vivado to correctly evaluate the timing of your circuit, you must put a register barrier at all inputs and outputs. It is good practice to put these barriers in a separate entity (a wrapper entity).



Multiple Clock Domains

- All the Assigned projects are solvable with a **single clock domain**.
- Handling multiple clock domain within the same circuit is an advanced skill:
 - Clock Domain Crossing (CDC)
- If you choose to exploit multiple clocks within your design, take care of:
 - Physical interfacing of clocks.
 - Proper timing constraints.
- If you decide to use multiple clocks, you are strongly encouraged to discuss your project with us before the final submission of the assignment.

How to write a Report

The following few pieces of advice on how to write the final report for your assignment:

- Think of the assignment as a contract between You (The designer) and Us (The client) and read carefully the assignment text: it will explicitly ask for some information that needs to be present in the report (e.g. Maximum Clock Frequency, Analysis on the critical path, etc...)
- A good report should be easy to read: the easier is for a reviewer to understand your work, the higher will be your evaluation of it.
- A good report shall always contain: I) an introduction II) a brief report on the state of the art III) technical sections where in our case the focus is on the block diagram architecture, the HDL implementation, its verification and its implementation on FPGA IV) A conclusion section which properly summarize the work done analysing the most relevant findings.

How to write a Report

The following few pieces of advice on how to write the final report for your assignment:

- You should number each figure, table and equation to reference them in the text by their name.
- Tables and figures shall always be accompanied by a caption describing them
- Whenever you use material not written or drawn by you, you should properly reference the original source
- Figures shall be relevant to the text: inserting a screenshot that is impossible to read is pointless. It is OK to use screenshots rather than custom-drawn figures but be sure that they are easy to understand for the reader.

How to write a Report

The following few pieces of advice on how to write the final report for your assignment:

- You are asked to attach the code (compressed folder), therefore there is no need to populate the report with all the VHDL written: just insert the most relevant fragments you want to comment on.
- Prefer quantitative assertions rather than qualitative (e.g., “The circuits occupy few logic resources” shall be substituted by “the circuit occupies X% of logical resources available)
- Be aware of the claims you make, do not assert concepts or draw conclusions that you did not demonstrate and are not supported by facts
- You are requested to provide a test plan, not supposed to be exhaustive but accurate enough: be sure to test regular operation, reset conditions and corner cases

Report's Evaluation

- The report is given a mark (from 0 to 33), and it will contribute approximately 30% to your final exam mark.
- The Assignment is optional: if you carry it out and discuss your report, the VHDL and FPGA design flow will not be part of your final exam, as the assignment will cover them.
- If you do not wish to carry out the assignment or you are not satisfied with the mark achieved, you will also be asked about this part of the course program during the exam.

Report's Evaluation

The evaluation will take into account:

- Quality of the HDL Code
 - Syntax, presence of latches, reset policies, appropriate use of constants and variables, quality of the code (clarity, comments)
- Verification of the system
 - Quality of the proposed test plan and its implementation; discussion of the results
- Synthesis & Implementation
 - Constraints definition, warning detailed analysis, max frequency, critical path discussion, resource utilization, power dissipation
- Overall Report's quality
 - A detailed introduction, circuit's block diagram, and overall clarity (remember, quality is not quantity!)
- Quality and Timeliness of Presentation

Report's Evaluation

- Discussion will be done in presence at the department facility
- 15 minutes – tight – to present work done.
- No need for dedicated ppt, report is enough
- Then approximately 15 minutes discussion
- LLMs and AI-assisted design:
 - Strong suggestion to avoid use them: project is optional, the point is to learn and get experience.
 - AI generated text in report is not tolerated – just be concise and don't worry on English grammar
 - During the discussion, you will be asked to justify the choices made and the way you wrote your code and you will be evaluated not only on the code you wrote, but also on the explanations you will give on your choices.
 - Fraction of code AI-generated are tolerated, just disclose what you generated with AI and be ready to justify the content of each line, code will be threaten as if it was written by you.
 - Code and report will be automatically checked for AI-generated content.

Project Hand off

- Send Project folder via email (See example work_env uploaded on teams)
- Object of the e-mail: [ES25 Project] – Your Name
- Send email to luca.fanucci@unipi.it, pietro.nannipieri@unipi.it and tommaso.bocchi@phd.unipi.it
- At least 3 working days BEFORE oral exam

End, Questions?

